



TOTALITY

A generator of hypnotic Berlin techno

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About

Totality composes, mixes and plays hypnotic Berlin techno: tracks of six to eight minutes and DJ sets of up to four hours, in which the tracks follow one another on two decks as a DJ would mix them. Nothing is played back from a recording. The kick, the rumble under it, the twelve lanes of the drum kit, the ping, the bass and the 303, the dub chord, the drone and the texture are synthesised; the tape echo, the spring, the plate and the rooms are simulated; and every note, every operation of the form and every move of a knob is decided by the composer.

The music it aims at is the one of Berghain and Ostgut Ton, of Dub Techno and of the hypnotic school: a groove that holds for minutes and changes by one element at a time, a low end made of a kick and the rumble of its own reverb, a chord thrown into a tape echo. Four style profiles span it -- Hypnotic in the centre, Ostgut, Dub and Raw -- and every knob of their recipes was fitted against measurements of thirty reference tracks: their spectral balance, their width, their loudness and its range, and how alike their bars are.

The same seed always gives the same track. A track is a seed, a style, a length and a handful of rerolls, which is what a set file stores and what you can hand to someone else. The standalone and the plugin play what the offline renderer plays, sample for sample.

Totality is the moment of a total eclipse when the moon covers the sun entirely and only the corona shines round the dark disc. In Totality's Patterns view the kick is that disc: every other part turns around it in rings, and its corona swells with every beat.

Quick start

The standalone plays by itself: press Play. Compose writes a new track with the knobs as they stand, New seed draws a fresh one. The status line tells what plays: its style and form, its tempo, its key with the Camelot label a DJ mixes by, and its length. While a track begins to play, Totality measures its loudest part in the background and brings it to its style's loudness; the status says "levelling" until the correction has glided in.

For a set, move Set min above zero: Totality then writes a set of that many minutes -- a dramaturgy of tempo and energy, tracks in neighbouring keys, blends of 16 or 32 bars with the bass swapped on the 32-bar line, loops of the last track under the next on the third deck, breaks from the mixer's effects.

In a DAW, Totality is an instrument plugin that follows the host's transport: it plays when the host plays, jumps when the host jumps, and plays at the host's tempo (a set's own tempo drift is the standalone's and the export's). Its output is a finished stereo mix at about -10 LUFS with its true peak under -1 dBTP.

A reroll draws one part of the track again and leaves the rest as it was: form, harmony, rack, layers, blocks, events, hands, sounds. In a set the rerolls act on the track under the playhead, and "reroll the whole track" or "the set's plan" draw more at once (the Arrange page).

The track and the set



A track has one of three forms. The Arc is the DJ tool: an intro of 32 or 64 bars from the kick on, a body that adds one layer every 32-bar block, an outro that takes them away again -- so it mixes. The Peak builds to a long kick-out and returns into its densest block. The Endless is full from the first bar and changes only by exchanging layers, as the hypnotic records do. Every block has exactly one operation (add, remove, swap) on its first bar -- a hat or a perc may come in on the block's 8- or 16-bar line instead --, every change falls on a 4-bar line and a return on a 16-bar line; nothing tonal plays in the first and last 32 bars, so two tracks can be mixed without a clash.

Every track has a figure: the voice it is remembered by -- a ping motif for Hypnotic, a chord stab or a bass riff for Ostgut, the dub chord for Dub, a 303 line for Raw. It plays the same one or two bars from its entry (the body's second block; a bass riff with the body) to the end and changes only through its filters. Between the operations the body runs in waves of 32 or 64 bars: the figure's filters and the hats' bus open towards a wave's end and fall back after it; two to four bars before that landing something drops away -- the centre of the mix, the kick and the claps, the figure -- so that the landing lands; in the middle of a wave one element (the figure, the bass or the rumble, the percussion, the open hats and the ride) breathes out for 8 or 16 bars and comes back with an echo throw. The Arrange page names the figure and the bars where the waves land.

Style picks the profile; Morph To and Morph blend it with a second one; the Dub and Hypnotic shares pull any profile along the two axes the style is described by. With Auto the tempo, the key and the owner of the low end are drawn from the profile; without it the knobs decide. Minutes is the track's length, rounded to whole blocks.

A set is mixed the way the Berghain residents describe their nights: as one long recomposition, not a row of crossfades. Each track is heard for about three minutes of its own (Track Time) and a third of it lies under its

neighbours -- about twenty tracks an hour. The incoming track comes in with its intro, its highs first and its mids over the last 16 bars, its low band killed; on a 32-bar line the bass goes over in one move; the outgoing track keeps its hats a while, loses its mids and highs and fades. The third deck borrows: the hats and percussion of the track that just left carry on under the new one, the next track's figure is teased in, high-passed, before its blend (where the keys agree), or a percussion loop of the track before last is layered under the one playing. Between the blends the DJ's hand works the channel on the 16-bar lines -- a low kill before the line so the kick slams back, the highs taken out and faded back in, a dip of the mids, the high-pass filter drawn up into the line, an echo throw on the last beat -- never where the track itself already drops something.

The set knobs: its length, the dramaturgy (Warm-up, Peak, Closing, Sunday, Flat -- the arc of tempo and energy over its length), whether the styles wander with the energy or stay, how often the third deck borrows (Live Loops), how often a break comes from the mixer's effects, the blend's length, a track's own time (Track Time) and how busy the DJ's hand is (DJ Hand). The mixer's effects are one send per deck into a tape echo and a long hall, which the set's breaks throw into and the Perform page's throw uses.

COMPOSE

Control	Key	Range	Default
Tempo	<code>compose.bpm</code>	100 .. 160 BPM	130 BPM
Key	<code>compose.key</code>	C, C#, D, D#, E, F, F#, G, G#, A, A#, B	A
Scale	<code>compose.scale</code>	Aeolian, Dorian, Phrygian, Hexachord, Minor Pentatonic	Aeolian
Style	<code>compose.style</code>	Hypnotic, Ostgut, Dub, Raw Peak	Hypnotic
Length	<code>compose.minutes</code>	2 .. 16 min	7 min
Swing	<code>compose.swing</code>	50 .. 66 %	53 %
Humanize	<code>compose.humanize</code>	0 .. 10 ms	3 ms
Low End	<code>compose.low_owner</code>	Rumble, Sub	Rumble
Auto	<code>compose.auto</code>	off / on	on
Form	<code>compose.form</code>	Auto, Arc, Peak, Endless	Auto
Morph To	<code>compose.morph_to</code>	Off, Hypnotic, Ostgut, Dub, Raw Peak	Off
Morph	<code>compose.morph</code>	0 .. 1	0.5
Dub Share	<code>compose.dub_share</code>	0 .. 1	0
Hypnotic Share	<code>compose.hypnotic_share</code>	0 .. 1	0
Composer's Sounds	<code>compose.pick_sounds</code>	off / on	on

SET

Control	Key	Range	Default
Dramaturgy	<code>set.dramaturgy</code>	Warm-up, Peak, Closing, Sunday, Flat	Peak
Styles	<code>set.journey</code>	Stay, Wander	Wander
Live Loops	<code>set.loops</code>	0 .. 1	0.5
FX Breaks	<code>set.fx_breaks</code>	0 .. 1	0.3
Blend	<code>set.blend</code>	16 bars, 32 bars	32 bars
Set Length	<code>set.minutes</code>	0 .. 240 min	0 min
Track Time	<code>set.track_minutes</code>	2 .. 6 min	3 min

DJ Hand	set.dj_hand	0 .. 1	0.5
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DJFX

Control	Key	Range	Default
Echo Time	djfx.echo_time	1/16, 1/8, 3/16, 1/4, 3/8, 1/2	3/16
Feedback	djfx.feedback	0 .. 1	0.45
Echo Return	djfx.echo_return	-60 .. 6 dB	-3 dB
Hall Decay	djfx.hall_decay	0.5 .. 12 s	5 s
Hall Return	djfx.hall_return	-60 .. 12 dB	6 dB

Sounds and presets

Every synth has 1024 factory presets: the kick, the rumble, the sub, a lane of the kit, the ping, the bass, the 303, the dub chord, the drone and the texture. Each engine has sixteen groups -- "Berlin 909", "Dub Sine", "Ride Bell", "Classic 303", "Basic Channel" -- and each group sixty-four presets on an eight by eight grid: the rows an adjective from dark to bright ("Muffled" ... "Cutting"), the columns a noun of the group's own, so the name tells where on the grid a sound lies. A preset is the sound, not the mix: levels, pans, sends, ducking, the octaves and a lane's role and pattern stay where the mix and the composer put them.

For every track the composer chooses a preset per synth and per lane of the kit (Composer's Sounds, `compose.pick_sounds`): a group by how well it suits the track's style, a lane only among the presets made for its role, and in the group a row that leans to the middle -- the darkest and the brightest are chosen less often. The style then holds a few knobs in its own range (the Dub kick's soft click, the Raw kick's hard clip, the Dub chord's bright band). Reroll sounds draws other presets and leaves the notes as they are.

The knobs show what plays. When a track begins -- in a set, when the next track starts on its deck -- its presets and its style's mix are put on the knobs, and each page names the preset the composer chose ("this track: Punchy Drum Hall (Hypnotic Thud)"); the Arrange page lists them for the whole track. Turn a knob and the sound follows from there; choose another preset from the menu, or step through them, and it plays at once. In a blend each deck keeps its own track's sound: the knobs show the incoming track while the outgoing one plays on as it was.

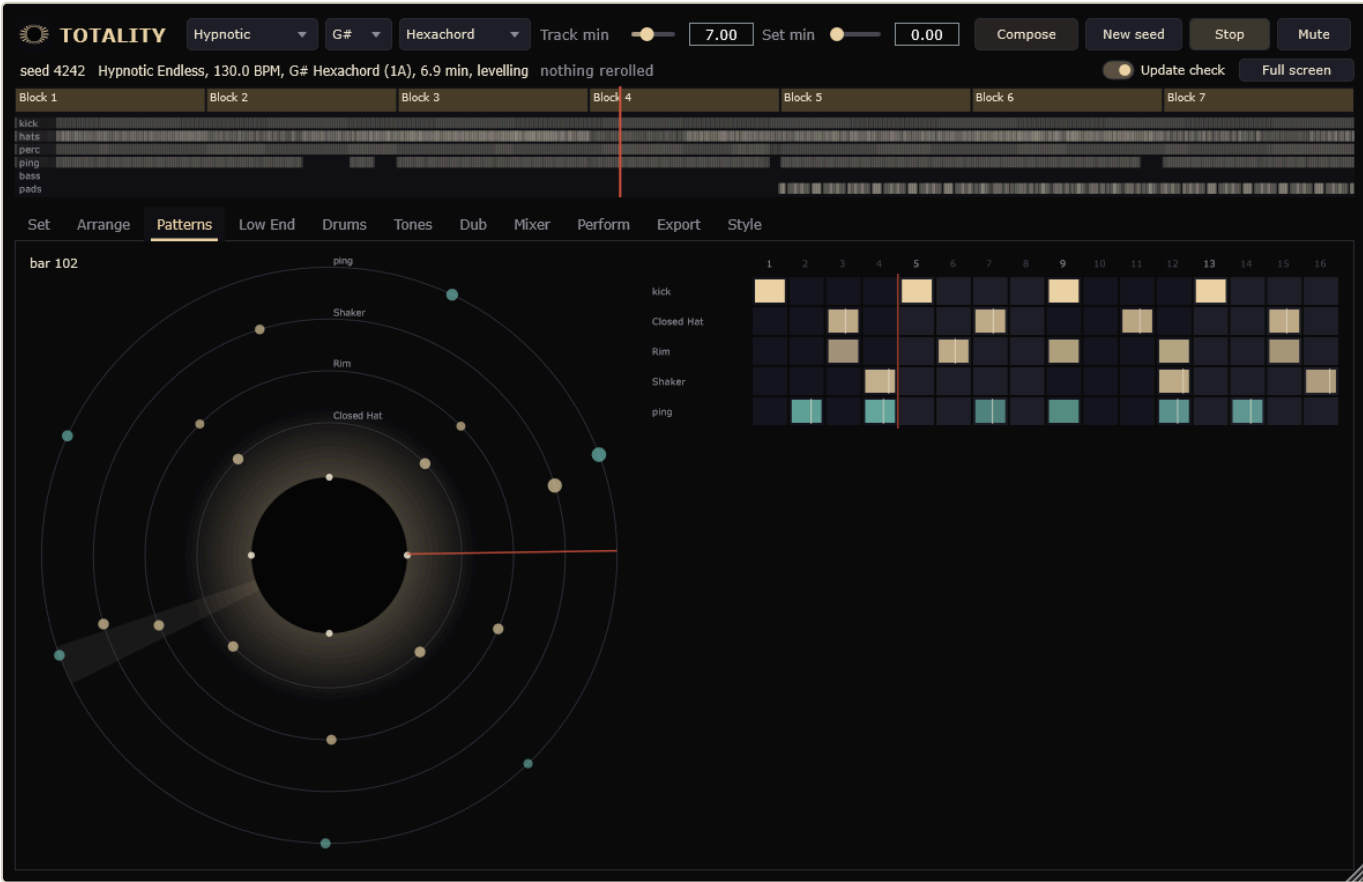
Arrange

The screenshot displays the 'TOTALITY' software interface in the 'Arrange' view. At the top, the track is named 'Hypnotic' with a key signature of 'G#' and a hexachord of 'Hexachord'. The tempo is set to 130.0 BPM, and the duration is 6.9 minutes. The interface shows a sequence of 7 blocks. A vertical red line indicates the current position in the arrangement. Below the blocks, there is a section for rerolling various elements: form, harmony, rack, layers, blocks, events, hands, sounds, and figure. The bottom part of the interface shows a detailed piano roll for the selected block, with lanes for kick, hats, perc, ping, bass, and pads. The piano roll shows the timing and pitch of the notes for each instrument.

The arrangement across the whole length: in a set the tracks on decks A and B (their swaps marked: from there a track owns the low end), in a track its blocks with their names, and under them a lane per group of layers -- kick, hats, perc, ping, bass, pads -- lit where it plays. The Arrange page shows the operations of the form by name. A click jumps there.

The row of rerolls acts on the track under the playhead; each part of it is drawn from a stream of its own, so a reroll of the blocks keeps the harmony and the sounds, a reroll of the hands (the automation) keeps every note, and a reroll of the figure draws another motif for the same voice.

Patterns and the Eclipse



The kick is the dark disc in the middle, and every other part that plays in the bar is a ring around it: a bar once round, its onsets as beads, larger the louder, the playhead's hand sweeping over them. A layer on the sixteen-step matrix shows the same beads bar after bar; a polyme-ter or a slipping layer moves on from bar to bar -- it precesses, and you can see it. Where three rings or more have an onset on the same sixteenth, the corona lights up at that angle: a conjunction. Beside it the same bar as a step grid, with a tick where an onset lies off the grid (swing and feel).

The screenshot shows the TOTALITY software interface, specifically the 'Low End' system settings. The interface is dark-themed with various knobs and sliders for configuring the kick, rumble, and sub bass. The 'Low End' tab is selected, showing settings for the kick, rumble, and sub bass. The kick settings include parameters like Punch, Punch Decay, Attack, Hold, Decay, Tail Limit, Click, Click Tone, Click Decay, Tone, Low Cut, Dip Freq, Dip, Top Level, Top Pitch, Top Decay, Top Drive, Top Cut, Rumble, Rumble Level, Rumble Split, Rumble Drive, Rumble Resonance, Room, Room Decay, Room Size, Room Damping, Room Pre-Delay, Room Low Pass, Sub, Sub Level, Sub Octave, Sub Drive, Sub Low Pass, Sub Kick Lock, Envelope, Envelope Attack, Envelope Decay, Envelope Sustain, Envelope Release, and Duck, Duck Duck, Duck Hold, Duck Release.

The low end is one system. The kick hands its phase on: the rumble's sub continues the kick's tail at the phase the kick left it, and the sub bass locks its phase to the last kick, so the three never cancel each other. The kick

has three engines (a swept sine, a resonator, a modelled 909) and a top layer; its drive runs at four times the rate.

The rumble is the kick sent into a hall, cut below the split, clipped and ducked by the next kick -- the dark roll of Berlin techno. Either the rumble or the sub bass owns the band under 80 Hz (compose.low_owner): never both at once.

KICK

Control	Key	Range	Default
Engine	kick.engine	Sweep, Resonator, 909	Sweep
Tune	kick.tune	Free, Key, Fifth, Flat Seventh	Key
Pitch End	kick.pitch_end	30 .. 120 Hz	52 Hz
Pitch Start	kick.pitch_start	60 .. 1500 Hz	300 Hz
Body Decay	kick.pitch_decay	5 .. 300 ms	18 ms
Punch Decay	kick.punch_decay	0.5 .. 20 ms	3 ms
Punch	kick.punch	0 .. 1	0.45
Attack	kick.amp_attack	0 .. 10 ms	0.2 ms
Hold	kick.amp_hold	0 .. 150 ms	25 ms
Decay	kick.amp_decay	20 .. 1500 ms	380 ms
Drive	kick.drive	0 .. 1	0.35
Clip	kick.clip	Soft, Hard	Soft
Click	kick.click_level	0 .. 1	0.35
Click Tone	kick.click_tone	500 .. 12000 Hz	3000 Hz
Click Decay	kick.click_decay	0.5 .. 30 ms	6 ms
Tone	kick.tone	200 .. 20000 Hz	6000 Hz
Level	kick.level	-36 .. 6 dB	-3 dB
Tail Limit	kick.tail_limit	-60 .. 0 dB	0 dB
Low Cut	kick.low_cut	15 .. 60 Hz	30 Hz
Dip Freq	kick.dip_freq	200 .. 1500 Hz	500 Hz
Dip	kick.dip	-12 .. 0 dB	-3 dB
Top Level	kick.top_level	-60 .. 0 dB	-12 dB
Top Pitch	kick.top_pitch	-24 .. 0 st	-8 st
Top Decay	kick.top_decay	20 .. 400 ms	90 ms
Top Drive	kick.top_drive	0 .. 1	0.6
Top Cut	kick.top_cut	150 .. 2000 Hz	400 Hz

RUMBLE

Control	Key	Range	Default
Level	rumble.level	-60 .. 6 dB	-9 dB
Split	rumble.split	50 .. 150 Hz	80 Hz
Decay	rumble.decay	0.5 .. 6 s	1.8 s

Size	rumble.size	0.3 .. 3	1
Damping	rumble.damping	0 .. 1	0.5
Pre-Delay	rumble.predelay	0 .. 60 ms	0 ms
Drive	rumble.drive	0 .. 18 dB	6 dB
Low Pass	rumble.ratio	1.5 .. 6 x f0	3 x f0
Resonance	rumble.resonance	0 .. 1	0.15
Sub	rumble.sub	-60 .. 6 dB	0 dB
Sub Attack	rumble.sub_attack	5 .. 200 ms	25 ms
Sub Release	rumble.sub_release	50 .. 2000 ms	300 ms
Duck	rumble.duck	0 .. 24 dB	7 dB
Duck Hold	rumble.duck_hold	0 .. 250 ms	80 ms
Duck Release	rumble.duck_release	30 .. 600 ms	220 ms

SUB

Control	Key	Range	Default
Level	sub.level	-60 .. 6 dB	-6 dB
Octave	sub.octave	-2 .. 2	0
Attack	sub.attack	0.5 .. 50 ms	3 ms
Decay	sub.decay	20 .. 2000 ms	250 ms
Sustain	sub.sustain	0 .. 1	0.7
Release	sub.release	10 .. 800 ms	90 ms
Low Pass	sub.low_pass	40 .. 300 Hz	110 Hz
Drive	sub.drive	0 .. 1	0.1
Kick Lock	sub.lock	Off, Kick	Kick
Duck	sub.duck	0 .. 24 dB	9 dB
Duck Hold	sub.duck_hold	0 .. 250 ms	40 ms
Duck Release	sub.duck_release	30 .. 600 ms	180 ms

Drums



Twelve lanes, each a small drum synthesiser: a role (the closed and rolling hats, the open hat, the ride, the claps, the snare, the rim, the shaker, toms and congas, noise), an engine (noise, metal -- the 909's six square waves -- modal, tone, FM), a filter, a shape. They run side by side in the lanes of the processor's vector registers. The hats and the percussion each have a bus with its level and a low pass, then the drum bus's saturation; the ducks let the pads and the rooms step aside for the kick, band by band.

PERC (12 INSTANCES)

Control	Key	Range	Default
Active	percN.active	off / on	on
Role	percN.role	Closed Hat, Rolling Hat, Open Hat, Ride, Clap, Clap Ghost, Snare, Rim, Shaker, Tom, Conga, Noise	Closed Hat
Engine	percN.engine	Noise, Metal, Modal, Tone, FM	Noise
Pitch	percN.pitch	40 .. 12000 Hz	400 Hz
Pitch Amount	percN.pitch_amount	1 .. 16 x	1 x
Pitch Decay	percN.pitch_decay	0.5 .. 300 ms	10 ms
FM Ratio	percN.fm_ratio	0.25 .. 8	1.41
FM Index	percN.fm_index	0 .. 8	0
Modes	percN.mode_set	Membrane, Bar, Harmonic	Membrane

Mode Damping	percN.mode_damp	0 .. 1	0.5
Metal Scale	percN.metal_scale	0.25 .. 4 x	1 x
Noise	percN.noise	0 .. 1	1
Noise Decay	percN.noise_decay	2 .. 3000 ms	70 ms
Bursts	percN.bursts	1 .. 6	1
Burst Spacing	percN.burst_spacing	2 .. 40 ms	10 ms
Decay	percN.decay	2 .. 3000 ms	70 ms
Filter	percN.filter	Low Pass, Band Pass, High Pass	High Pass
Cutoff	percN.cutoff	100 .. 18000 Hz	7000 Hz
Resonance	percN.resonance	0 .. 1	0.15
Low Cut	percN.low_cut	150 .. 8000 Hz	3000 Hz
Drive	percN.drive	0 .. 1	0.1
Level	percN.level	-36 .. 6 dB	-0.7 dB
Pan	percN.pan	-1 .. 1	0.1
Choke Group	percN.choke	0 .. 4	1
Shift	percN.shift	-10 .. 10 ms	0 ms
Density	percN.density	0 .. 1	0.5
Tune to Key	percN.tune	off / on	off
Pan Depth	percN.pan_depth	0 .. 1	0.3
Pan Period	percN.panBars	0.0625 .. 16 bars	0.1875 bars
Cut Tracks Pitch	percN.cut_track	0 .. 2	0
Noise Type	percN.noise_type	White, 909 Metal	909 Metal

MIX

Control	Key	Range	Default
Hats Level	mix.hats_level	-24 .. 12 dB	-3.5 dB
Hats Cut	mix.hats_cut	200 .. 20000 Hz	20000 Hz
Perc Level	mix.perc_level	-24 .. 12 dB	-6.5 dB
Perc Cut	mix.perc_cut	200 .. 20000 Hz	20000 Hz
Drum Saturation	mix.drum_sat	0 .. 1	0.35
Duck Low	mix.duck_low	0 .. 18 dB	10 dB
Duck Mid	mix.duck_mid	0 .. 12 dB	3 dB
Group Low Cut	mix.low_cut	20 .. 500 Hz	20 Hz

Tones

The ping is the hypnotic school's figure: FM through a low-pass gate, a band pass that wanders. The bass and the 303 share one synthesiser with ten circuit filters (the Moog ladder, the 303's diode ladder, the Prophet's SSM2040, the Juno's IR3109, the SEM, the MS-20's Korg35 and more); the 303 line has its accents and slides. The dub chord is a minor seventh or a stab of six voices through a band pass, a crusher and a phaser, sent into the tape echo and the plate. The drone holds a note under eight bars; the texture is vinyl crackle, hum and erosion.

PING

Control	Key	Range	Default
Level	ping.level	-60 .. 6 dB	-12.5 dB
Pan	ping.pan	-1 .. 1	0
Width	ping.width	0 .. 1	0.5
Ratio	ping.ratio	0.25 .. 8	1.414
Index	ping.index	0 .. 6	2
Index Decay	ping.index_decay	2 .. 500 ms	60 ms
Pitch Amount	ping.pitch_amount	0 .. 4	0.8
Pitch Decay	ping.pitch_decay	2 .. 300 ms	45 ms
Decay	ping.decay	10 .. 2000 ms	140 ms
Low-Pass Gate	ping.lpg	0 .. 1	0.6
Gate Release	ping.lpg_release	10 .. 2000 ms	150 ms
Resonance	ping.resonance	0 .. 1	0.35
Band	ping.band	150 .. 6000 Hz	900 Hz
Band Q	ping.band_q	0.2 .. 8	0.8
Band Mix	ping.band_mix	0 .. 1	0.5
Sweep	ping.sweep	0 .. 2 oct	0.6 oct
Sweep Rate	ping.sweep_rate	0.01 .. 2 Hz	0.07 Hz

BASS

Control	Key	Range	Default
Level	bass.level	-60 .. 6 dB	-9 dB
Pan	bass.pan	-1 .. 1	0
Wave	bass.wave	0 .. 1	0
Pulse Width	bass.pulse_width	0.05 .. 0.95	0.5
Sub Osc	bass.sub_osc	0 .. 1	0.3
Filter	bass.filter	Moog Ladder, Prophet (SSM2040), Juno (IR3109), Oberheim SEM, Xpander, Diode Ladder (303), Korg35 (MS-20), Polivoks, EDP Wasp, Comb	Juno (IR3109)
Cutoff	bass.cutoff	40 .. 12000 Hz	350 Hz
Resonance	bass.resonance	0 .. 1	0.2
Env Amount	bass.env_amount	0 .. 6 oct	2 oct
Decay	bass.decay	5 .. 3000 ms	180 ms

Accent	bass.accent	0 .. 1	0.3
Attack	bass.amp_attack	0.5 .. 500 ms	3 ms
Amp Decay	bass.amp_decay	5 .. 3000 ms	250 ms
Sustain	bass.amp_sustain	0 .. 1	0.6
Release	bass.amp_release	5 .. 3000 ms	300 ms
Glide	bass.glide	0 .. 500 ms	0 ms
Drive	bass.drive	0 .. 1	0.2
Key Track	bass.key_track	0 .. 1	0.5
Low Cut	bass.low_cut	20 .. 500 Hz	65 Hz
High Cut	bass.high_cut	500 .. 20000 Hz	5000 Hz
Dub Send	bass.dub_send	0 .. 1	0
Room Send	bass.room_send	0 .. 1	0
Duck	bass.duck	0 .. 24 dB	8 dB
Duck Release	bass.duck_release	30 .. 600 ms	200 ms

ACID

Control	Key	Range	Default
Level	acid.level	-60 .. 6 dB	-14 dB
Pan	acid.pan	-1 .. 1	0.1
Wave	acid.wave	0 .. 1	0
Pulse Width	acid.pulse_width	0.05 .. 0.95	0.5
Sub Osc	acid.sub_osc	0 .. 1	0
Filter	acid.filter	Moog Ladder, Prophet (SSM2040), Juno (IR3109), Oberheim SEM, Xpander, Diode Ladder (303), Korg35 (MS-20), Polivoks, EDP Wasp, Comb	Diode Ladder (303)
Cutoff	acid.cutoff	40 .. 12000 Hz	600 Hz
Resonance	acid.resonance	0 .. 1	0.72
Env Amount	acid.env_amount	0 .. 6 oct	2.5 oct
Decay	acid.decay	5 .. 3000 ms	220 ms
Accent	acid.accent	0 .. 1	0.6
Attack	acid.amp_attack	0.5 .. 500 ms	3 ms
Amp Decay	acid.amp_decay	5 .. 3000 ms	250 ms
Sustain	acid.amp_sustain	0 .. 1	0.8
Release	acid.amp_release	5 .. 3000 ms	60 ms
Glide	acid.glide	0 .. 500 ms	60 ms
Drive	acid.drive	0 .. 1	0.5
Key Track	acid.key_track	0 .. 1	0.3
Low Cut	acid.low_cut	20 .. 500 Hz	120 Hz
High Cut	acid.high_cut	500 .. 20000 Hz	9000 Hz

Dub Send	acid.dub_send	0 .. 1	0.25
Room Send	acid.room_send	0 .. 1	0.01
Duck	acid.duck	0 .. 24 dB	4 dB
Duck Release	acid.duck_release	30 .. 600 ms	200 ms

CHORD

Control	Key	Range	Default
Level	chord.level	-60 .. 6 dB	-5 dB
Width	chord.width	0 .. 1	0.6
Detune	chord.detune	0 .. 30 ct	10 ct
Octave	chord.octave	-2 .. 2	0
Attack	chord.attack	0.5 .. 500 ms	10 ms
Decay	chord.decay	20 .. 3000 ms	300 ms
Sustain	chord.sustain	0 .. 1	0.15
Release	chord.release	10 .. 3000 ms	400 ms
Brightness	chord.bright	200 .. 12000 Hz	2500 Hz
Env Amount	chord.env_amount	0 .. 5 oct	1.5 oct
Band	chord.band	100 .. 3000 Hz	380 Hz
Band Mode	chord.band_mode	Band Pass, High Pass	Band Pass
Band Q	chord.band_q	0.2 .. 4	0.7
Sweep	chord.sweep	0 .. 2 oct	0.5 oct
Sweep Rate	chord.sweep_rate	0.01 .. 2 Hz	0.2 Hz
Bits	chord.crush	4 .. 16	8
Crush Mix	chord.crush_mix	0 .. 1	0.2
Phaser	chord.phaser	0 .. 1	0.5
Phaser Rate	chord.phaser_rate	0.05 .. 4 Hz	0.5 Hz
Dip 600 Hz	chord.dip	-18 .. 0 dB	-7 dB
Dub Send	chord.dub_send	0 .. 1	0.5
Plate Send	chord.plate_send	0 .. 1	0.35

DRONE

Control	Key	Range	Default
Level	drone.level	-60 .. 6 dB	-13 dB
Octave	drone.octave	-2 .. 2	0
Detune	drone.detune	0 .. 30 ct	8 ct
Cutoff	drone.cutoff	80 .. 8000 Hz	700 Hz
Resonance	drone.resonance	0 .. 1	0.25
Sweep	drone.sweep	0 .. 4 oct	1.5 oct
Sweep Bars	drone.sweep_bars	4 .. 128 bars	32 bars

Attack	drone.attack	0.01 .. 20 s	4 s
Release	drone.release	0.01 .. 20 s	6 s
Plate Send	drone.plate_send	0 .. 1	0.4
Room Send	drone.room_send	0 .. 1	0.02

TEXTURE

Control	Key	Range	Default
Level	texture.level	-60 .. 0 dB	-26 dB
Crackle	texture.crackle	0 .. 1	0.5
Hum	texture.hum	0 .. 1	0.2
Hum Freq	texture.hum_hz	40 .. 70 Hz	50 Hz
Erosion	texture.erosion	0 .. 1	0.4
Width	texture.width	0 .. 1	0.8

Dub



The dub chain: a tape echo with wow, flutter and drive, a spring, a plate. The ping, the chord and the kit's buses send into it, and the composer throws single hits into the echo on the 8-bar lines. The room is the hall the hats, the percussion and the ping live in; the grain cloud hears the ping and the chord and scatters them.

DUB

Control	Key	Range	Default
Echo Time	dub.echo_time	1/16, 1/8, 3/16, 1/4, 3/8, 1/2	3/16
Feedback	dub.feedback	0 .. 1.2	0.25
Echo Tone	dub.tone	800 .. 12000 Hz	4500 Hz
Echo Low Cut	dub.low_cut	40 .. 800 Hz	200 Hz
Wow	dub.wow	0 .. 3 ms	0.5 ms
Flutter	dub.flutter	0 .. 0.5 ms	0.05 ms
Echo Drive	dub.drive	0 .. 18 dB	4 dB
Echo Return	dub.echo_return	-60 .. 6 dB	-3 dB
Spring Return	dub.spring	-60 .. 6 dB	-9 dB
Spring Decay	dub.spring_decay	0.3 .. 6 s	2 s

Plate Decay	dub.plate_decay	0.5 .. 12 s	4.2 s
Plate Damping	dub.plate_damping	0 .. 1	0.4
Plate Pre-Delay	dub.plate_predelay	0 .. 100 ms	20 ms
Plate Low Cut	dub.plate_low_cut	60 .. 800 Hz	250 Hz
Plate Return	dub.plate_return	-60 .. 18 dB	6 dB
Ping to Echo	dub.ping_send	0 .. 1	0.12
Hats to Echo	dub.hats_send	0 .. 1	0
Perc to Echo	dub.perc_send	0 .. 1	0

SPACE

Control	Key	Range	Default
Return	space.level	-60 .. 12 dB	0 dB
Size	space.size	0.3 .. 3	0.9
Decay	space.decay	0.2 .. 12 s	1.2 s
Damping	space.damping	0 .. 1	0.5
Pre-Delay	space.predelay	0 .. 100 ms	15 ms
Low Cut	space.low_cut	60 .. 800 Hz	250 Hz
High Cut	space.high_cut	1000 .. 18000 Hz	6000 Hz
Hats Send	space.hats_send	0 .. 1	0.25
Perc Send	space.perc_send	0 .. 1	0.3
Ping Send	space.ping_send	0 .. 1	0.4

CLOUD

Control	Key	Range	Default
Level	cloud.level	-60 .. 6 dB	-60 dB
Density	cloud.density	0.5 .. 60 /s	12 /s
Grain Size	cloud.size	20 .. 1000 ms	180 ms
Pitch	cloud.pitch	0 .. 1	0.3
Spray	cloud.spray	0.05 .. 4.5 s	2 s
From Ping	cloud.ping_send	0 .. 1	0.8
From Chord	cloud.chord_send	0 .. 1	0.6
Plate Send	cloud.plate_send	0 .. 1	0.5

Mixer and Master



The meters: each deck after its mixer channel, the output, and the loudness of the last 400 ms. Each deck's track bus has a group high pass (the form's filter sweeps), a tilt of the highs against the lows, and the glue -- a bus compressor mixed in at 35 per cent -- after which the loudness correction sits. The master keeps the side under Mono Below mono, sets the level, has an optional vinyl cut, a clipper at four times the rate and a true-peak limiter.

The motion is the free-running modulation of the whole instrument: the hats' cutoff and level, their decay, the percussion's cutoff, the rumble's drive, slow LFOs of prime periods that never line up.

A deck's channel is a DJ mixer's: the fader, a three-band isolator (200 Hz and 2.5 kHz, Linkwitz-Riley, each band from a kill to +6 dB), a filter that is a low pass to the left and a high pass to the right, and the send into the mixer's effects. In a set the composer moves them as a DJ would; in the plugin they are yours too.

MIX

Control	Key	Range	Default
Hats Level	<code>mix.hats_level</code>	-24 .. 12 dB	-3.5 dB
Hats Cut	<code>mix.hats_cut</code>	200 .. 20000 Hz	20000 Hz
Perc Level	<code>mix.perc_level</code>	-24 .. 12 dB	-6.5 dB
Perc Cut	<code>mix.perc_cut</code>	200 .. 20000 Hz	20000 Hz
Drum Saturation	<code>mix.drum_sat</code>	0 .. 1	0.35
Duck Low	<code>mix.duck_low</code>	0 .. 18 dB	10 dB
Duck Mid	<code>mix.duck_mid</code>	0 .. 12 dB	3 dB
Group Low Cut	<code>mix.low_cut</code>	20 .. 500 Hz	20 Hz

MASTER

Control	Key	Range	Default
Level	<code>master.level</code>	-24 .. 12 dB	0 dB
Glue Threshold	<code>master.threshold</code>	-40 .. 0 dB	-16 dB
Glue Ratio	<code>master.ratio</code>	1 .. 10	2
Clip Drive	<code>master.clip</code>	0 .. 9 dB	2 dB
Ceiling	<code>master.ceiling</code>	-6 .. 0 dBTP	-1 dBTP
Mono Below	<code>master.mono_below</code>	40 .. 250 Hz	120 Hz
Tilt	<code>master.tilt</code>	-12 .. 12 dB	10 dB
Vinyl Cut	<code>master.cut</code>	off / on	off

MOTION

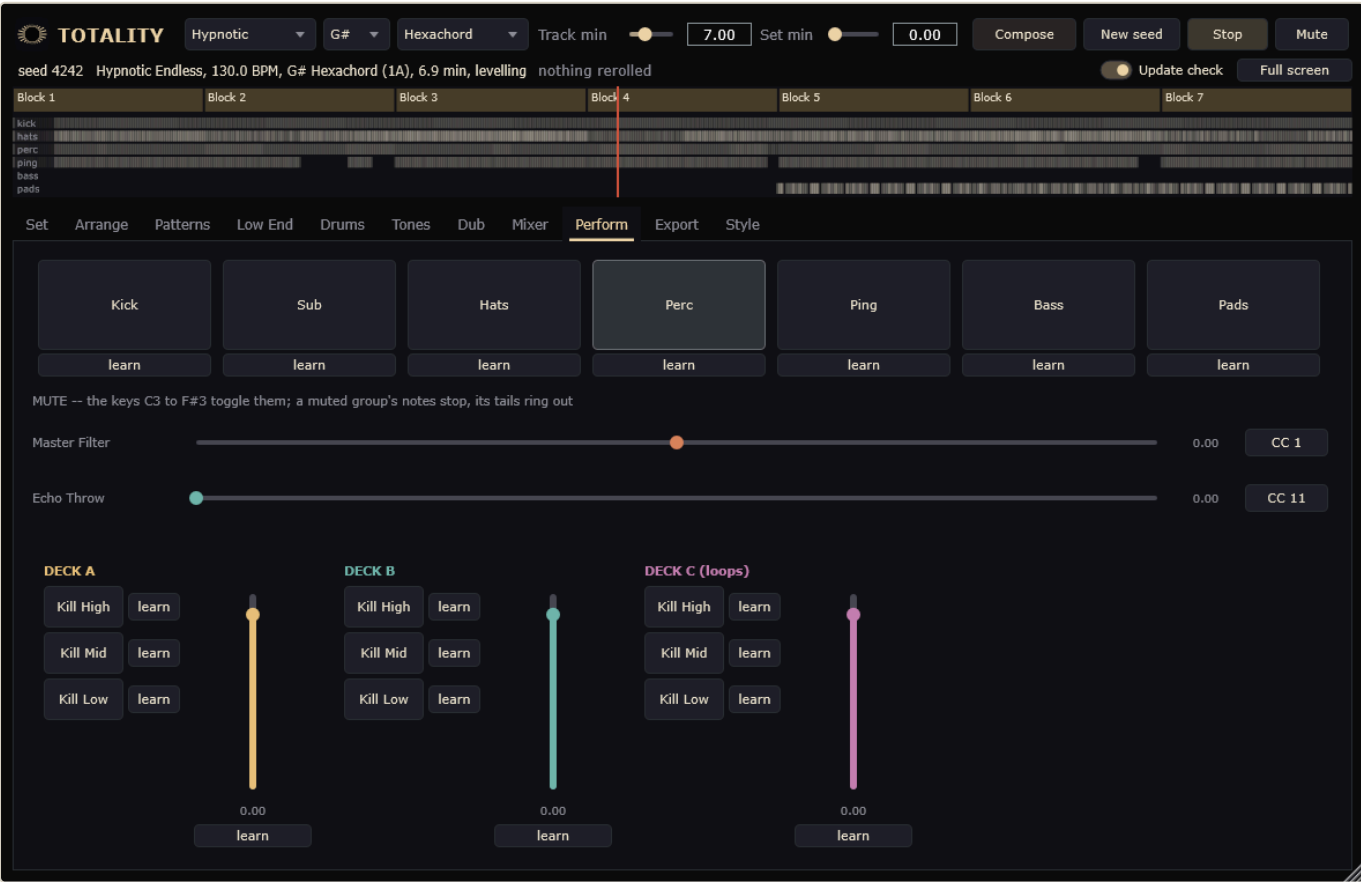
Control	Key	Range	Default
Amount	<code>motion.amount</code>	0 .. 1	1
Hats Cut	<code>motion.hats_cut</code>	0 .. 1.5 oct	0.4 oct
Hats Level	<code>motion.hats_level</code>	0 .. 6 dB	1.5 dB
Hat Decay	<code>motion.hat_decay</code>	0 .. 0.5	0.2
Perc Cut	<code>motion.perc_cut</code>	0 .. 1.5 oct	0.4 oct
Rumble Drive	<code>motion.rumble_drive</code>	0 .. 6 dB	2 dB

DECK (3 INSTANCES)

Control	Key	Range	Default
Fader	<code>deckN.fader</code>	-60 .. 6 dB	0 dB
Low	<code>deckN.low</code>	-60 .. 6 dB	0 dB
Mid	<code>deckN.mid</code>	-60 .. 6 dB	0 dB

High	deckN.high	-60 .. 6 dB	0 dB
Filter	deckN.filter	-1 .. 1	0
FX Send	deckN.fx_send	0 .. 1	0

Perform



As at a mixer. Seven mutes: a muted group's notes stop and its tails ring out -- the kick out, the classic move. The master filter takes the whole mix, the echo throw sends it into the mixer's tape echo. Each deck has its kills and its fader. Every control can be learned from a MIDI controller: click learn, move the controller (a right click forgets it). By default the mod wheel moves the filter and the expression pedal the throw, and the keys from middle C up (C to F#) toggle the mutes of kick, sub, hats, perc, ping, bass and pads. The bindings are saved with the project.

The perform controls are live only: an export or tot_render plays the track as it was composed.

PERFORM

Control	Key	Range	Default
Master Filter	perform.filter	-1 .. 1	0
Echo Throw	perform.throw	0 .. 1	0
Mute Kick	perform.mute_kick	off / on	off
Mute Sub	perform.mute_sub	off / on	off
Mute Hats	perform.mute_hats	off / on	off
Mute Perc	perform.mute_perc	off / on	off
Mute Ping	perform.mute_ping	off / on	off
Mute Bass	perform.mute_bass	off / on	off
Mute Pads	perform.mute_pads	off / on	off

TOTALITY

Hypnotic ▾

G# ▾

Hexachord ▾

Track min 7.00

Set min 0.00

Compose

New seed

Stop

Mute

seed 4242 Hypnotic Endless, 130.0 BPM, G# Hexachord (1A), 6.9 min, levelling nothing rerolled

☒ Update check

Full screen

Block 1

Block 2

Block 3

Block 4

Block 5

Block 6

Block 7

kick

hats

perc

ping

bass

pads

Set

Arrange

Patterns

Low End

Drums

Tones

Dub

Mixer

Perform

Export

Style

WAV + MIDI + cues

... with stems

... with DJ loops

... with both

Save .totset

Load .totset

seed 4242 Hypnotic Endless, 130.0 BPM, G# Hexachord (1A), 6.9 min, levelling

The export renders what plays as it was composed (the performer's mutes, filter and throw are live only) at 48 kHz:

- a 24-bit WAV with its cues as markers (the bass's entry, the kick-outs and returns, the outro; in a set every track, swap and loop) and the same cues as JSON beside it (<name>.wav.cues.json),
- the MIDI file with the tempo map and a channel per part,
- with stems: a 32-bit WAV per element into <name>_stems; their sum is the mix before the master, exactly,
- with DJ loops (a track): 4 and 8 bars of its loudest block, seamless, the mix and kick, hats and perc alone.

An .totset holds the seed, the lengths, the rerolls and every changed knob.

OSC CUES

OSC Cues

OSC Port

☐

9000

A set of 40 minutes on the Arrange page: nine tracks on two decks, their blends, the swaps.

The export renders what plays at 48 kHz: a 24-bit WAV with cue markers (the bass's entry, the kick-outs and returns, the outro; in a set every track, swap and loop) and the same cues as JSON beside it, for DJ software that does not read the WAV's markers; the MIDI file with the tempo map and a channel per part; on request the stems -- a WAV per element, whose sum is exactly the mix before the master -- and the DJ loops of a track: 4 and 8 bars of its loudest block, seamless, as the mix and as kick, hats and percussion alone. An .totset file holds the seed, the lengths, the rerolls and every changed knob.

The OSC cues tell a visualiser such as Kaleidoscope what the score knows, at the moment it is heard: the beat and the tempo, the bar, the block's name, the form's operation, the key. They go to this machine at the port given, or to TOT_CUE_HOST.

```
/tot/beat i f      beat number, tempo
/tot/bar i        bar number
/tot/block s      the block's name ("Intro", "Block 3", "Reduction", a set's track)
/tot/op s         the operation ("add ride", "kick out")
/tot/key s        the key as a Camelot label ("8A")
```

CUE

Control	Key	Range	Default
OSC Cues	cue.enabled	off / on	off
OSC Port	cue.port	1024 .. 65535	9000

Style



The numbers of the profile the knobs describe, and the references' measurements per style that the profiles are fitted to. A style of your own: Take the profile's numbers, change what you like, switch Use on -- the tempo range, the weights of the three forms, the length, the rack's mutation and reroll, polymeters, fills, swing, the filtered edges, the longest kick-out, the events, the corridor of bar similarity the composer's candidates are chosen against, and the loudness the Leveler brings each track to.

CUSTOM

Control	Key	Range	Default
Use Custom Style	custom.use	off / on	off
Tempo Low	custom.bpm_low	120 .. 140 BPM	128 BPM
Tempo High	custom.bpm_high	120 .. 140 BPM	133 BPM
Arc Weight	custom.arc	0 .. 1	0.5
Peak Weight	custom.peak	0 .. 1	0.2
Endless Weight	custom.endless	0 .. 1	0.3
Sub Owns Low	custom.sub_chance	0 .. 1	0.3
Blocks Low	custom.blocks_low	4 .. 12	6
Blocks High	custom.blocks_high	4 .. 12	8
Mutation	custom.mutation	0 .. 1	0.25
Reroll	custom.reroll	0 .. 1	0.15

Polymer	custom.polymer	0 .. 1	0.8
Fills	custom.fill	0 .. 1	0.15
Filtered Edges	custom.edge	0 .. 1	0.5
Longest Kick-out	custom.max_reduction	4 .. 32 bars	32 bars
Swing Low	custom.swing_low	50 .. 62 %	51 %
Swing High	custom.swing_high	50 .. 62 %	56 %
Events	custom.event_rate	0 .. 1	0.35
Throw Share	custom.throw_share	0 .. 1	0.5
Density Cap	custom.density_cap	3 .. 12	9
Bar Similarity	custom.similarity	0.7 .. 1	0.9
Target LUFS	custom.peak_lufs	-14 .. -7 LUFS	-10 LUFS

Meta Quest

The whole generator runs on the headset (build-quest\TotalityQuest.apk, see Quest/README.md). The hands are the hands at the mixer: left pinch play and stop, right pinch kick out and in, both hands together the next track; the left hand's height the master filter, the right's the echo throw. Above and ahead of the player the Eclipse turns at the size of a room. A file tot.cfg in the app's folder sets the seed, the style, a set's length and any knob.

The command line

tot_render composes and renders without a window, the same samples the plugin plays:

```
tot_render --seed 7 --out track.wav --midi track.mid --stems stems --loops loops
tot_render --seed 2026 --set-minutes 120 --out set.wav --score-json set.json
tot_render --set "compose.style=Dub; compose.minutes=8" --seed 3 --out dub.wav
tot_render --list                every parameter with its value
tot_render --bench --quality quest how fast, at the headset's quality
```

The evaluation (Tools/eval_report.py) measures rendered tracks as the references were measured and writes a report of what lies outside their corridor.